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ORIGINAL ARTICLE

The impact of AI applications, information sharing, and supply chain resilience on agricultural supply chain performance

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Abstract

In the digital era, improving supply chain performance (SCP) has become increasingly important for agricultural enterprises confronting global competition and market volatility. These organizations need to manage networks comprising multiple stakeholders, diverse information flows, and complex logistics processes, while addressing unique challenges in the agricultural sector. This study examines the interrelationships among four latent variables using partial least squares structural equation modeling (PLS-SEM): artificial intelligence (AI) applications, information sharing (IS), and supply chain resilience (SCR) as independent variables, with SCP as the dependent variable. Six hypotheses were tested using data collected from 151 agricultural enterprises in China through a structured questionnaire. This study contributes to the literature on supply chain management by examining how the triad of AI applications, IS, and SCR influences agricultural SCP. It provides insights into potential approaches for enterprises to implement AI-driven solutions in addressing sector-specific challenges like perishability management and multi-stakeholder coordination in volatile markets.

Keywords Agricultural supply chain · Performance · AI applications · Information sharing · Resilience · PLS-SEM

1 Introduction

An agricultural supply chain (ASC) represents a complex network of organizations engaged in upstream and downstream processes, which encompasses the production, processing, and distribution of agricultural products to end consumers (Joel et al. 2024). Modern ASCs face unique challenges in managing operations such as farming, harvesting, processing, storage, and distribution, primarily due to the inherent characteristics of agricultural products, including their perishable nature and seasonal variability (Oluwafunmi Adijat Elufioye et al., 2024). To address these challenges, AI applications have shown potential as valuable tools to enhance the efficiency of ASC (Dhal 2024). These applications, which consist of intelligent systems, machine learning algorithms, and advanced analytics tools, enable data-driven decision-making, predictive modeling, and automated operational management in various supply chain activities (Helo and Hao 2022).

The strategic deployment of AI applications has become increasingly important for ASCs seeking to enhance their operational efficiency and competitive advantage (Javier et al. 2024). Empirical evidence suggests that specialized AI applications can provide notable benefits across supply chain networks, particularly in yield prediction, inventory optimization, and automated quality control systems (Zhao et al. 2023). These AI applications demonstrate versatility in both vertical integration (from farmers to retailers) and horizontal coordination (across similar operational levels), enabling smart farming practices and intelligent distribution networks (Ganeshkumar et al. 2023).

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Performance assessment in ASCs requires particular attention due to the unique challenges of product perishability and seasonal variations (Zhang and Li 2024).

Research indicates that supply chain performance (SCP) is significantly influenced by multiple factors, including the level of sophistication of AI applications, the effectiveness of information sharing (IS) mechanisms between partners, and supply chain resilience (SCR) capabilities (Shahzadi et al. 2024; Yin et al. 2025). The benefits of implementing comprehensive performance evaluation systems in ASCs are multifaceted, including improved resource utilization, enhanced quality control, more accurate demand forecasting, and strengthened data-driven decision-making capabilities. Furthermore, while technology adoption presents certain implementation challenges, proper performance measurement systems help organizations identify and address these barriers while maximizing the benefits of digital transformation (Bhilat et al. 2024).

Contemporary literature employs sophisticated SEMs to examine the determinants and measurement of agricultural SCP. Recent empirical studies have emphasized the growing focus on technological adoption and digital transformation impacts in performance measurement frameworks (Zhang and Li 2024). Through comprehensive SEM analyses, Ali et al. (2024) identified significant relationships among AI application effectiveness, IS capabilities, and SCR in agricultural systems. Their findings demonstrate that the implementation of AI applications and digital collaboration enhance agricultural SCP through improved operational coordination. The integration of agriculture 4.0 technologies has shown potential to enhance competitive advantages through improved operational efficiency and enhanced resilience capabilities (Mehmood 2024). Furthermore, empirical evidence suggests that investments in technological infrastructure and digital capability development contribute significantly to SCR enhancement and overall performance outcomes (Farrukh Shahzad et al. 2025). Kumar and Raj (2024) further establish technological readiness and AI adoption as critical success factors in ASC operations.

Although previous studies have examined AI applications, IS and SCR independently, there remains a research gap in understanding their interrelated effects specifically within ASC (Panigrahi 2024). ASCs face distinct challenges stemming from their inherent characteristics: seasonality of production, product perishability, weather dependence, and complex stakeholder networks from farm to table (Arowosegbe et al. 2024; Zhang 2024). These characteristics require effective coordination mechanisms, including real-time information exchange and adaptive decision-making capabilities throughout the supply chain network. This study contributes to addressing this gap by empirically examining how these factors can interact to influence SCP in China's agricultural sector, offering insights into AI-driven and resilience-focused optimization.

The primary objectives of this study are to:

- Develop a conceptual framework to examine these relationships in the agricultural context.
- Examine the Impact of AI applications, IS, and SCR on agricultural SCP.

2 Literature review and hypothesis development

2.1 Dynamic capabilities theory

This study employs dynamic capabilities theory (DCT) as the theoretical framework to conceptualize the relationships between AI applications, IS, SCR, and SCP in agricultural contexts. DCT emphasizes a firm's capacity to adapt to environmental changes by reconfiguring resources and processes to maintain competitive advantage (Liu 2022; Teece 2018). The theory defines dynamic capabilities as the ability to integrate, build, and reconfigure internal and external competences to address rapidly changing environments through three core processes: sensing, seizing, and reconfiguring (Teece 2009).

SCR represents an outcome of dynamic capabilities, specifically, the organisation's ability to anticipate, respond to, and recover from disruptions (Ponomarov and Holcomb 2009). In agricultural contexts, AI applications operationalize sensing capabilities by enabling predictive analytics and pattern recognition for early disruption detection (Belhadi et al. 2024), while IS mechanisms actualize seizing capabilities through real-time coordination and information exchange among supply chain partners (García-Alcaraz et al. 2021). Together, these technologies support reconfiguring capabilities by enabling flexible resource allocation and adaptive responses. SCP emerges as the combined result of these dynamic capabilities and their effective implementation across the supply chain network.

Recent studies have expanded its application in diverse contexts. Ahmed (2024) critiques alternative perspectives of the theory, highlighting debates on its tautological nature and proposing criteria for defining organizational dynamic capabilities. Furthermore, van de Wetering (2021) explores dynamic capabilities driven by enterprise architecture, revealing their role in enhancing organizational agility through digital project benefits. Additionally, case studies in Brazilian agribusiness illustrate the alignment of dynamic capabilities with internationalization strategies (Tondolo et al., 2010). These advances underscore the theory's adaptability to technological, environmental, and strategic challenges in both developed and emerging economies.

The perishable nature of agricultural products intensifies the need for dynamic capabilities. For example, AI-driven cold chain monitoring systems leverage real-time IS to dynamically adjust storage conditions, integrating sensing, seizing, and reconfiguring capabilities to reduce spoilage and improve performance (Dhal 2025).

2.2 AI applications

AI applications constitute sophisticated technological solutions that enable computer systems to emulate human cognitive abilities through advanced processing capabilities, including machine learning algorithms, natural language processing systems, and autonomous reasoning mechanisms (Joel et al. 2024). In modern supply chain management (SCM), AI applications have evolved into fundamental transformative tools, encompassing predictive analytics, machine learning algorithms, automated decision-making systems, and intelligent forecasting mechanisms (Parra-Hernández et al. 2024). These AI-powered capabilities are projected to become increasingly central to supply chain operations by 2025, particularly in pattern recognition, autonomous planning, and advanced predictive modeling (Khedr and S 2024).

Implementing AI applications in supply chains represents a strategic integration process in which intelligent systems are systematically incorporated to improve operational efficiency and decision-making capabilities. This transformation involves the deployment of specific AI applications at multiple supply chain stages, enabling organizations to optimize operations through intelligent automation and advanced analytics (Ganeshkumar et al. 2023). In ASCs, specifically, AI applications serve as a comprehensive approach to resource optimization and cost reduction, effectively addressing complex challenges in demand forecasting, inventory management, and quality control, particularly in situations characterized by uncertainty and variability (Belhadi et al. 2024).

2.3 IS

IS in ASCs has evolved significantly with the integration of modern technologies. The success of agricultural products depends critically on efficient information flow across supply chain nodes, facilitated by Internet of Things (IoT) sensors, cloud computing systems, and integrated management platforms (Raji 2024). These technologies enable systematic sharing of critical data on crop conditions, harvest timing, storage conditions, and transportation requirements, creating more responsive and efficient supply chain networks (Sharma and Shivandu 2024).

The effectiveness of IS is particularly crucial in agricultural contexts due to the perishable nature of products and seasonal production patterns (García-Alcaraz et al. 2021). Key aspects include real-time crop condition monitoring, storage environment tracking, and transportation condition management (He 2024). Although implementing comprehensive IS systems requires significant technological investment, the benefits often outweigh the costs through improved inventory management and reduced waste (Kalimuthu 2024). Organizations that effectively implement IS systems can better coordinate activities and make more informed decisions about production and distribution (Obonyo et al. 2025).

However, there are several challenges in the implementation of agricultural IS. Primary concerns include data security and privacy issues, which must be carefully balanced against the benefits of transparency (Shoomal et al. 2024). Additionally, non-systematic IS can lead to market instabilities and operational inefficiencies. Appropriate data protection mechanisms and structured IS protocols require careful implementation to maintain competitive advantages while ensuring system security (Kusolchoo 2024).

Within ASCs, AI applications constitute a significant advance in information management and operational efficiency. These intelligent systems demonstrate remarkable capabilities in processing large amounts of data and generating actionable insights through advanced analytics and machine learning algorithms (Rathore et al. 2021). The strategic integration of AI applications with complementary technologies such as IoT and blockchain has enabled real-time monitoring and improved decision-making capabilities throughout the supply chain (Dhal 2024).

Furthermore, AI applications have shown particular effectiveness in improving IS through improved prediction and response capabilities. Implementing distributed AI applications enables autonomous decision-making and real-time optimization, while also facilitating pattern identification and trend prediction, leading to more efficient and sustainable supply chain operations (Sharifmousavi et al. 2024). Therefore,

> H1:AI applications have a direct and positive effect on IS in ASCs.

2.4 SCR

In agricultural contexts, SCR involves multiple dimensions, including the ability to anticipate potential disruptions, respond efficiently to disturbances, and adapt to changing conditions while maintaining continuous operations (Ojo

2024). Recent research emphasizes that agricultural SCR must encompass both proactive and reactive capabilities, allowing supply chains not only to recover from disruptions but also to transform and improve their operational resilience through adaptive mechanisms (Parra-Hernández et al. 2024).

Given these complexities, resilience planning is particularly crucial due to the distinctive challenges posed by weather uncertainties, seasonal variations, and product perishability (Zhao et al. 2024). Empirical research has identified key resilience strategies, including supplier diversification, the development of flexible transportation networks, the implementation of adaptive inventory management systems and establishment of robust contingency planning mechanisms (Yuan et al. 2024).

The ability to process vast amounts of historical and real-time data allows ASCs to develop more effective contingency plans and respond more quickly to unexpected events (Trollman 2024). This is particularly evident in how AI-driven systems help organizations not only predict potential disruptions but also automatically initiate appropriate response mechanisms, thus significantly improving overall SCR (Belhadi et al. 2024). Hence,

H2: AI applications have a direct and positive effect on SCR in ASCs.

Similarly, IS plays a vital role in the development of SCR through multiple mechanisms. It enables supply chain partners to maintain operational visibility and coordinate responses effectively during disruptions (Co 2024). The exchange of critical information on market conditions and potential risks allows ASC members to develop more robust response strategies (Zhong et al. 2024).

Furthermore, the quality and effectiveness of IS directly impact the ability of supply chain partners to make informed decisions and maintain operational continuity (Ali et al. 2024). This is particularly crucial in agricultural contexts, where coordinated information exchange helps mitigate risks and improve overall SCP (Shukla et al. 2023). Accordingly,

H3: IS has a direct and positive effect on SCR in ASCs.

2.5 SCP

SCP in agricultural contexts requires a comprehensive measurement approach that takes into account multiple dimensions. The perishable nature of agricultural products makes quality maintenance and delivery timeliness particularly crucial performance indicators (Akram 2023). Research shows that effective performance measurement must consider both efficiency (cost management) and effectiveness (customer satisfaction) metrics (Marhamati 2023).

The seasonal nature of agricultural production adds complexity to performance measurement, requiring supply chain managers to account for varying production patterns and demand fluctuations (Khazaeli et al. 2024). Modern performance measurement systems must balance multiple objectives, including quality maintenance, cost efficiency, and customer satisfaction, while considering the unique challenges of perishable agricultural products (Yamsa-ard 2024).

Advanced AI applications have proven their effectiveness in improving decision-making processes and operational efficiency through advanced analytics and automation capabilities (Khedr and S 2024). Strategic implementation of AI applications leads to measurable improvements in cost reduction and resource allocation efficiency (Bhilat et al. 2024). Furthermore, the impact of AI applications on SCP is particularly evident in demand forecasting and inventory management, where predictive analytics help optimize operations and reduce waste (Oluwafunmi Adijat Elufioye et al., 2024). The enhanced capability of AI applications to process complex data patterns allows better quality control and customer satisfaction through improved service delivery and resource use (Adenekan et al. 2024). As a result,

H4: AI applications have a direct and positive effect on SCP in ASCs.

IS plays a vital role in improving SCP in agricultural contexts. Research shows that effective IS leads to improved coordination among partners and significant reductions in operational uncertainties (Zhang and Li 2024). When ASC partners share accurate and timely information on inventory levels and production schedules, it results in optimized operations and improved overall performance (Sharma 2020).

Furthermore, evidence supports that IS contributes to better supply chain visibility and stakeholder coordination (Kalimuthu 2024). Implementing effective IS mechanisms has been shown to improve operational efficiency and enable better decision-making processes throughout the supply chain (Huo et al. 2021). Thus,

H5: IS has a direct and positive effect on SCP in ASCs.

In fact, SCR plays a crucial role in improving SCP in agricultural settings. Research shows that resilient supply chains demonstrate superior ability to maintain operational continuity and quickly recover from interruptions (Nguyen and Le 2024). Implementing resilience strategies has been proven to ensure reliable availability of resources and minimize the impact of supply chain disruptions (Muzamil et al. 2024).

Furthermore, evidence supports that resilience factors contribute significantly to improved planning strategies and stakeholder collaboration (Türkeş et al. 2024). The ability to quickly recover from disruptions and maintain performance levels during challenging conditions has been shown to be crucial for sustainable business operations and enhanced customer satisfaction (Kazancoglu et al. 2024). Subsequently,

H6: SCR has a direct and positive effect on SCP in ASCs (Fig. 1).

3 Methodology

3.1 Measures

The development of the questionnaire followed a systematic three-step approach. Initially, we developed a preliminary questionnaire through an extensive review of the literature to identify appropriate measurement scales for the key constructs. Subsequently, to enhance accessibility, we engaged language experts to translate our original English questionnaire into Chinese. To ensure the accuracy of translation, independent translators retranslated this Chinese version into English by independent translators. The two English versions were carefully compared to identify discrepancies or inconsistencies. Lastly, a pilot study was conducted with ten top-level executives. Based on feedback from the pilot test, we refined and modified the questionnaire to ensure its clarity and relevance to Chinese ASC practices.

According to the analysis in Sect. 3, the measurement scales were adopted from the established literature. IS was measured using 6 items from García-Alcaraz et al. (2021), while SCP was assessed through 9 items adapted from Lusiantoro et al. (2022). For measuring AI applications and SCR, 5 items each were adopted from Belhadi et al. (2024). Detailed items are listed in the appendix (Tables 5, 6,and 7). All measurements utilized a 7-point Likert scale. The modeling of structural equations was performed using Smart-PLS 4.0 software.

3.2 Sample and data collection

Following Ketokivi and Choi (2014), this research employed a quantified technique targeting CEOs and top executives in China's agricultural processing and retail enterprises from October 2023 to January 2024, using a prospective sampling approach (Gammelgaard 2017).

Research was carried out in a major agricultural enterprise group, labeled 'XYZ'. Established in the early 2000 s, the group operates more than 400 retail platforms in 75 major cities in China. The enterprise specializes in agricultural product processing, distribution, and retail operations, serving more than 300 million customers with fresh produce, pre-packaged agricultural products, and grain supplies. In each branch, we identified a key informant who was well versed in SCM practices. The questionnaires were initially distributed by email to 250 relevant experts. After sending reminder emails in 15 days, a second round of follow-up was conducted two weeks later.

Finally, we received 168 questionnaire responses and, after careful screening and data cleaning, 151 valid responses were retained, giving a response rate of 60.4%. Based on the analysis by G-power (f^2=0.15, α=0.05, predictors=4 and power=95%), this study requires a minimum sample size of 129, which is adequately met by this study.

Table 1 presents the demographic characteristics of the respondents.

Fig. 1 Proposed model

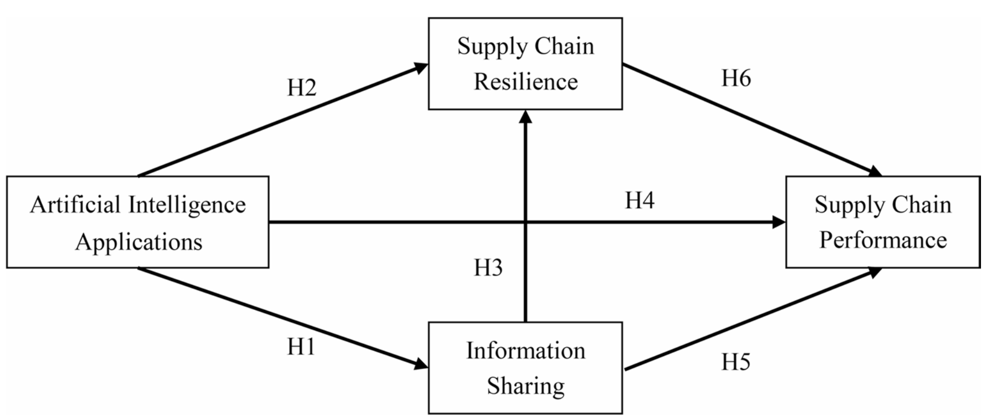

Table 1 Respondents' profile

Characteristic	Classification	Frequency	Percentage (%)
Role in the organization	CEO	98	64.90%
	Other top executives	53	35.09%
Business age (in year)	1–2	27	17.88%
	3–5	58	38.41%
	>=5	66	43.71%
Number of suppliers	1–5	66	43.71%
	6–10	47	31.13%
	>=10	38	25.17%
Branches	Processing	52	34.44%
	Distribution	38	25.17%
	Retail	45	29.80%
	Supporting Services	16	10.60%
Geographical distribution	Eastern China	68	45.03%
	Central China	53	35.10%
	Western China	30	19.87%

4 Results

4.1 Measurement model

We evaluated our measurement models through a comprehensive assessment of factor loadings, average variance extracted (AVE), and composite reliability (CR). As shown in Table 2, our measures demonstrate strong psychometric properties, with all factor loadings exceeding 0.75 (>0.6), AVE values ranging from 0.646 to 0.799 (exceeding the 0.5 threshold) and CR values between 0.908 and 0.937 (>0.7) (Hair et al. 2019). These results strongly support the reliability and convergent validity of our constructs. Furthermore, the variance inflation factor (VIF) values for all measures range from 1.962 to 4.154, with most values well below 5 (Hair et al. 2020), indicating that multicollinearity is not a significant concern in our measurement model.

Table 2 Convergent validity, reliability and collinearity

Constructs	Code	Loadings	AVE	CR	Cronbach's Alpha	VIF
Artificial Intelligence Applications (AIA)	AIA1	0.911	0.799	0.937	0.939	3.883
	AIA2	0.885				3.228
	AIA3	0.886				3.184
	AIA4	0.877				2.980
	AIA5	0.908				3.622
Information Sharing (IS)	IS1	0.896	0.700	0.914	0.927	3.243
	IS2	0.811				2.247
	IS3	0.823				2.322
	IS4	0.843				2.573
	IS5	0.842				2.466
	IS6	0.804				2.332
Supply Chian Resilience (SCR)	SCR1	0.908	0.731	0.908	0.919	3.276
	SCR2	0.850				2.475
	SCR3	0.834				2.418
	SCR4	0.825				2.243
	SCR5	0.855				2.396
Supply Chain Performance (SCP)	SCP1	0.795	0.646	0.931	0.937	2.492
	SCP2	0.900				4.154
	SCP3	0.756				2.030
	SCP4	0.770				2.412
	SCP5	0.842				2.889
	SCP6	0.795				2.426
	SCP7	0.765				1.962
	SCP8	0.805				2.355
	SCP9	0.795				2.304

Furthermore, Table 3 presents the heterotrait-monotrait ratio (HTMT) analysis, which shows all values ranging from 0.308 to 0.584, well below the conservative threshold of 0.85 (Franke and Sarstedt 2019). This confirms the discriminant validity among all constructs in the model.

4.2 Structural model

The evaluation of the structural model was carried out based on two primary criteria: the coefficient of determination (R^2) and the significance levels. R^2 values demonstrate the combined effect of exogenous variables on endogenous latent variables, effectively representing the percentage of variance in endogenous constructs that is explained by all connected exogenous constructs (Hair 2019). The results of our analysis of the structural model, including the variance explained for each construct, are illustrated in Fig. 2.

Table 3 Discriminant validity

Construct	AIA	IS	SCP	SCR
AIA				
IS	0.389			
SCP	0.584	0.384		
SCR	0.358	0.308	0.443	

AIA = Artificial Intelligence Applications, IS = Information Sharing, SCP = Supply Chain Performance, SCR = Supply Chain Resilience

Furthermore, the PLS analysis revealed that all latent variables had variance inflation factor (VIF) values below 5, confirming the absence of significant multicollinearity concerns. Specifically, the model goodness of fit indices (SRMR = 0.055 < 0.8) indicated an acceptable fit with the data (Hair 2022).

4.3 Path coefficient and hypotheses results

The results of the path model are presented in Fig. 2, and the detailed findings are summarized in Table 4.

Specifically, AI demonstrated strong and significant positive relationships with IS (β = 0.366, t = 4.943, p < 0.001), SCP (β = 0.422, t = 6.142, p < 0.001), and SCR (β = 0.267, t = 3.674, p < 0.001), supporting H1, H2 and H3, respectively. Although IS showed a significant positive effect on SCR (β = 0.191, t = 2.356, p < 0.05) supporting H5, its relationship with SCP was not significant (β = 0.140, t = 1.878, p > 0.05), therefore, H4 was not supported. Finally, the resilience of the supply chain demonstrated a significant positive effect on the performance of SCP (β = 0.235, t = 3.040, p < 0.01), supporting H6.

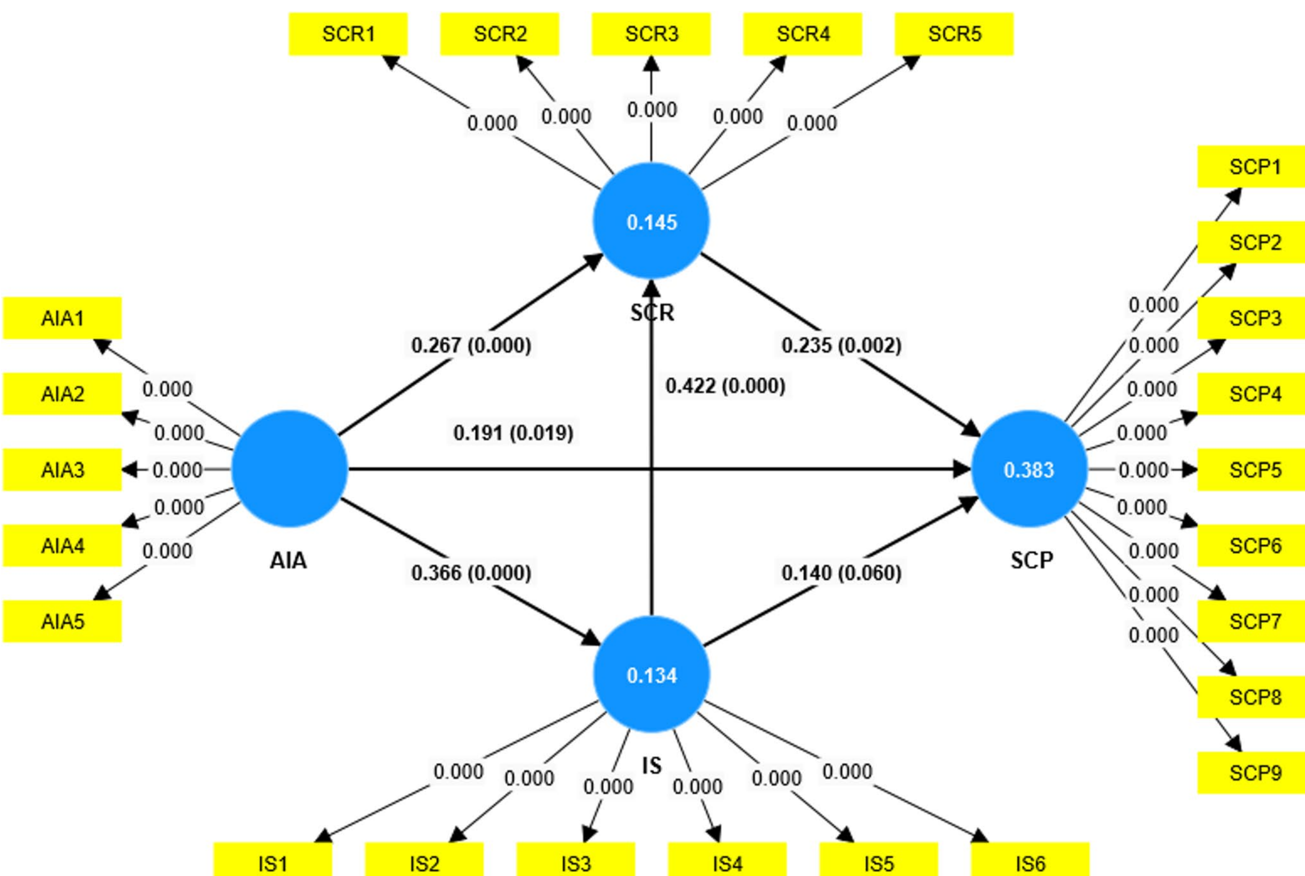

Fig. 2 Path model

Table 4 Path coefficients and hypotheses testing

Hypotheses	Relationship	β	t statistics	*p* values	Decision
H1	AIA ->IS	0.366	4.943	0.000	Supported
H2	AIA ->SCR	0.267	3.674	0.000	Supported
H3	IS ->SCR	0.191	2.356	0.019	Supported
H4	AIA ->SCP	0.422	6.142	0.000	Supported
H5	IS ->SCP	0.140	1.878	0.060	Not Supported
H6	SCR ->SCP	0.235	3.040	0.002	Supported

AIA = Artificial Intelligence Applications, *IS = Information Sharing*, *SCP = Supply Chain Performance*, *SCR = Supply Chain Resilience*

4.4 Discussion

The empirical evidence displayed in Fig. 2 demonstrates that five hypotheses were statistically supported while one was not supported, with Table 4 providing a summary of these conclusions.

Empirical results support H1, showing that AI applications positively impact IS (β=0.476, t=6.235, $p<0.001$), indicating that AI technologies serve as powerful enablers of improved IS practices, facilitating real-time data exchange, automated information processing, and intelligent data analytics between supply chain partners. The findings align with contemporary research (Lu et al. 2024) and demonstrate how AI-powered systems can effectively overcome traditional barriers to IS in agricultural contexts, such as data inconsistency, communication delays, and information asymmetry.

H2 was also supported, with the AI applications positively influencing SCR (β=0.398, t=4.892, $p<0.001$). This substantial effect indicates that AI technologies play a crucial role in building resilient ASCs through enhanced risk detection, predictive analytics, and automated response mechanisms. The findings extend previous research (Wang et al. 2025) by showing how AI-driven systems enable agricultural organizations to better anticipate disruptions, develop proactive mitigation strategies, and quickly adapt to changing conditions.

For H3, the results further validated the positive effect of IS on SCR (β=0.191, t=2.356, $p<0.05$), highlighting how effective information exchange mechanisms contribute to enhanced resilience in ASC. This finding aligns with previous research (Co 2024) and demonstrates that agricultural companies with robust IS practices are better positioned to identify risks, implement coordinated responses, and maintain operational continuity during disruptions.

The results also confirmed H4, with the AI applications having a significant impact on SCP (β=0.385, t=4.721, $p<0.001$). This substantial effect size indicates that AI technologies serve as a crucial driver for enhancing SCP through improved operational efficiency, better decision-making capabilities, and improved predictive analytics. The findings align with recent studies (Zhang and Li 2024; Shahzadi et al. 2024) and underscore how AI implementation enables agricultural companies to optimize inventory management, reduce waste of perishable goods, and improve demand forecast accuracy.

The non-significant direct relationship between IS and SCP in agricultural contexts (H5: β=0.082, $p>0.05$) appears to contradict some previous studies (Ali et al. 2024; Kumar 2024). Although IS improves SCR (β=0.191, $p<0.05$), which subsequently increases SCP (β=0.235, $p<0.01$), the perishability of agricultural products, the fragmented nature of stakeholder networks and the reliance on real-time adjustments may obscure the direct impact. Research on organic produce and Indian agri-chains confirms that IS primarily enhances SCP indirectly through resilience-building (e.g., risk anticipation through weather alerts) or AI-enabled automation (e.g., real-time logistics adjustments), rather than through standalone information exchange (Shukla et al. 2023).

The analysis strongly supports H6, confirming a significant positive relationship between SCR and SCP (β=0.423, t=5.128, $p<0.001$). This finding demonstrates that resilient ASCs are more capable of maintaining and improving their performance metrics. The size of the strong effect is consistent with recent literature (Yin et al. 2025) and emphasizes how resilience capabilities, including adaptability, flexibility, and rapid recovery from disruptions, directly contribute to improved operational efficiency and overall supply chain performance.

5 Implication

5.1 Theoretical implications

This study advances theoretical understanding by demonstrating how AI applications and IS function as dynamic capabilities under DCT, enabling agricultural firms to sense market opportunities, seize competitive advantages, and reconfigure resources to optimize perishable inventory management and mitigate seasonality risks through predictive analytics (Lima et al. 2025). It extends DCT by empirically demonstrating AI's role in the sensing dimension through disruption detection (e.g., weather-pattern prediction) and the reconfiguring dimension through adaptive logistics optimization (e.g., automated rerouting), enabling enhanced recovery capabilities compared to traditional approaches (Stadtfeld and Gruchmann 2024). The findings contribute to DCT by providing empirical evidence that AI-enhanced IS serves as a dynamic capability that transforms organizational routines, enabling firms to anticipate risks through machine learning-driven analytics rather than relying solely on reactive information processing (Tiwari et al. 2024). The non-significant direct IS-SCP relationship supports DCT's emphasis on capability orchestration, demonstrating that standalone information systems require dynamic capability integration—specifically AI enhancement—to generate sustainable competitive advantages and operational resilience in complex agricultural ecosystems (Lu et al. 2024).

5.2 Practical implications

Our findings offer four key actionable implications for ASC managers.

For strategic AI applications deployment as dynamic capabilities, managers should focus on developing sensing, seizing, and reconfiguring capabilities at critical vulnerability points in ASCs where perishability and seasonality create the greatest operational challenges (Nayal et al. 2022; Dora et al. 2022). A capability-building approach should begin with foundational sensing capabilities through high-impact applications such as AI-powered demand forecasting systems that detect market patterns, inventory optimization algorithms that sense stock level variations, and cold chain monitoring systems that identify temperature and quality deviations (Kumari et al. 2023). This phased development should then progress to seizing capabilities that capitalize on AI-generated insights through automated procurement decisions and resource allocation, before expanding to comprehensive reconfiguring capabilities that enable real-time operational adaptation and supply chain restructuring (Nayal et al. 2023).

For IS as capability infrastructure, our findings reveal that while IS does not directly enhance SCP, it serves as critical infrastructure for developing collective dynamic capabilities and enabling other performance drivers (Baah et al. 2022). Therefore, managers should therefore reframe IS initiatives from performance-focused to capability-enabling perspectives, recognizing that IS functions as a foundational infrastructure that supports sensing, seizing, and reconfiguring capabilities across supply chain networks (Panahifar et al. 2018). Rather than expecting direct performance returns, organizations should implement IS systems that enhance network-level sensing through collaborative risk identification and shared market intelligence, enable synchronized opportunity seizing through coordinated planning, and facilitate collective reconfiguring during disruptions (Sharma et al. 2024). Organizations with higher vulnerability to disruption should prioritize robust IT infrastructure for secure information exchange, addressing key ASC challenges of information asymmetry and unstable partnerships (Kumar and Kumar Singh 2022).

For SCR as a dynamic capability, our findings demonstrate that resilience functions as an adaptive capability comprising three sequential phases: readiness, response, and recovery, which work together to maintain operational continuity during ASC disruptions (Ali et al. 2024). Organizations should implement AI applications and IS mechanisms as key antecedents to the development of resilience capability, enhancing sensing, seizing, and reconfiguring capabilities for disruption detection, response mobilization, and recovery learning (Pal et al. 2024). The building of strategic resilience in ASCs should emphasize balanced configurations that address sector-specific challenges through digital technologies that enable real-time monitoring, automated responses, and predictive analytics for proactive management of disruption (Stadtfeld and Gruchmann 2024). The key imperative is to develop SCR as a dynamic capability that continuously evolves through learning, enabling flexible resource allocation and adaptive responses that ultimately improve SCP outcomes.

5.3 Implications of the policy

Policymakers should implement a comprehensive approach to support AI applications and the adoption of IS in ASCs through four key initiatives: establishing robust financial support mechanisms, including targeted subsidies and funding for AI-related initiatives (Mishra 2023); developing national strategies and regulatory frameworks for digital transformation that address agricultural production challenges (Dora et al. 2022); creating educational programs that unite farmers, agronomists, and supply chain experts to build technical capabilities (Cuong et al. 2024), and implement supportive infrastructure and governance frameworks that integrate AI and IoT technologies with appropriate data protection measures.

These integrated policy measures should focus on both technological advancement and the development of human capacity to ensure a successful digital transformation in the agricultural sector, while addressing implementation barriers and promoting the sustainable adoption of AI technologies in the agricultural supply chain.

6 Conclusion, limitations and directions for future research

This study empirically validates the critical role of AI applications, IS, and supply SCR in improving agricultural SCP. The results demonstrate that AI applications directly improve IS, SCR and SCP, underscoring their transformative potential to optimize decision making, risk mitigation, and operational efficiency. Although SCR significantly mediates performance results, IS exhibits an indirect influence through SCR rather than a direct effect on SCP, suggesting its role as an enabler of resilience rather than as a standalone performance driver.

There are some limitations. First, the narrow focus on quality and delivery metrics for SCP excludes important factors such as cost efficiency, resource utilization, and overall operational performance, limiting the completeness and applicability of the results. Second, the relatively small sample size of 151 agricultural enterprises can restrict the generalizability of the findings, as larger and more diverse samples could yield different insights. Furthermore, the geographical concentration of the research within China could influence the results due to regional-specific economic, cultural, and infrastructure factors, potentially limiting the applicability of these results to other global contexts. Finally, the reliance on self-reported data introduces potential biases related to the accuracy, honesty, and subjective interpretations of the respondents, which could affect the validity of the conclusions. Future research should address these limitations by employing larger and more diverse samples, broadening geographic scope to improve generalizability, incorporating objective data collection methods to mitigate self-report biases, and adopting more holistic SCP measurement frameworks that integrate additional operational performance dimensions.

Additionally, the scope of this study is limited to examining the effects of AI applications, IS, and SCR on SCP. Other scholars such as Zhang and Liu (2024) have investigated the role of digital twins and blockchain technology in agricultural SCP; Wang et al. (2023)

have analyzed the impact of sustainable practices and green innovation on ASC efficiency; and Kumar and Singh (2024) have explored the effects of IoT integration and cold chain management on agricultural SCP in emerging markets. This suggests that more research is needed in the context of ASC, given its unique characteristics such as product perishability, seasonal production patterns, weather dependencies, food safety requirements, small farmer integration, market price volatility, and rural infrastructure constraints. These aspects, along with emerging technologies such as precision agriculture, smart agriculture, and sustainable practices, present promising opportunities for future research.

Appendix

Table 5 Measurement items and sources

Constructs	Code	Loadings	Source
Artificial intelligence Application (AIA)	AIA1	The company has the infrastructure and expertise to implement AI-based information processing systems.	Belhadi et al. (2024)
	AIA2	The company uses artificial intelligence methods to anticipate and forecast environmental patterns.	
	AIA3	The company creates statistical models, self-learning algorithms, and predictive tools using artificial intelligence (AI) technologies.	
	AIA4	The company integrates AI techniques in all stages of the supply chain.	
	AIA5	The company shares AI-driven insights to support collaborative decision-making in the supply chain.	
Information Sharing (IS)	IS1	The company and its supply chain partners share essential data, such as demand forecasts.	García-Alcaraz et al. (2021)
	IS2	The company and its supply chain partners share information in a timely manner.	
	IS3	The company and its supply chain partners exchange precise and reliable information.	
	IS4	The company and its supply chain partners ensure the sharing of comprehensive information.	
	IS5	The company and its supply chain partners securely exchange sensitive and confidential information.	
	IS6	IS exists between the supply chain department and other departments.	
Supply Chian Resilience (SCR)	SCR1	Our supply chain has robust capabilities to address disruption-related challenges.	Belhadi et al. (2024)
	SCR2	Our supply chain shows agility in formulating and execution of emergency measures when disruptions occur.	
	SCR3	Our supply chain effectively manages unexpected interruptions through swift restoration of operational flows.	
	SCR4	Our supply chain exhibits rapid recovery capabilities, enabling prompt restoration to standard operational conditions after disruption.	
	SCR5	Our supply chain demonstrates transformative resilience by achieving higher levels of performance following disruptive events.	
Supply Chain Performance (SCP)	SCP1	The company ensures that customer orders are always received smoothly.	Lusiantoro et al. (2022)
	SCP2	The company ensures that the products are delivered accurately and without problems.	
	SCP3	The company guarantees that orders are fulfilled with the exact quantity requested.	
	SCP4	The company takes care to deliver the orders to the correct destination.	
	SCP5	The company provides clear and accurate documentation for payments.	
	SCP6	The company is committed to providing products of the highest quality.	
	SCP7	The company prioritizes the fast and reliable delivery of products.	
	SCP8	The company quickly and accurately resolves customer complaints.	
	SCP9	The company responds promptly and effectively to customer inquiries.	

Table 6 Descriptive statistics of measurement items

	Mean	Median	Standard deviation		Mean	Median	Standard deviation
AIA1	4.954	5	1.533	SCP3	4.974	5	1.548
AIA2	4.927	5	1.474	SCP4	4.967	5	1.592
AIA3	5.086	5	1.578	SCP5	4.94	5	1.604
AIA4	5.093	5	1.592	SCP6	4.848	5	1.602
AIA5	4.841	5	1.424	SCP7	4.96	5	1.611
IS1	4.755	5	1.668	SCP8	4.424	4	1.662
IS2	4.781	5	1.703	SCP9	4.497	4	1.683
IS3	4.828	5	1.556	SCR1	4.709	5	1.58
IS4	4.848	5	1.594	SCR2	5.212	5	1.564
IS5	4.921	5	1.555	SCR3	4.874	5	1.462
IS6	4.894	5	1.648	SCR4	4.788	5	1.555
SCP1	4.788	5	1.65	SCR5	4.907	5	1.541
SCP2	4.854	5	1.605				

Table 7 Correlation matrix of constructs

	AIA	IS	SCP	SCR
AIA	1	0.366	0.553	0.337
IS	0.366	1	0.363	0.289
SCP	0.553	0.363	1	0.418
SCR	0.337	0.289	0.418	1

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethical statement The authors declare that this manuscript is original, has not been published before, and is currently not considered for publication elsewhere. The research being reported in the manuscript has not been submitted simultaneously to any other journal. We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who meet the criteria for authorship but are not listed.

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